

ASGR Assignment 5 Questions

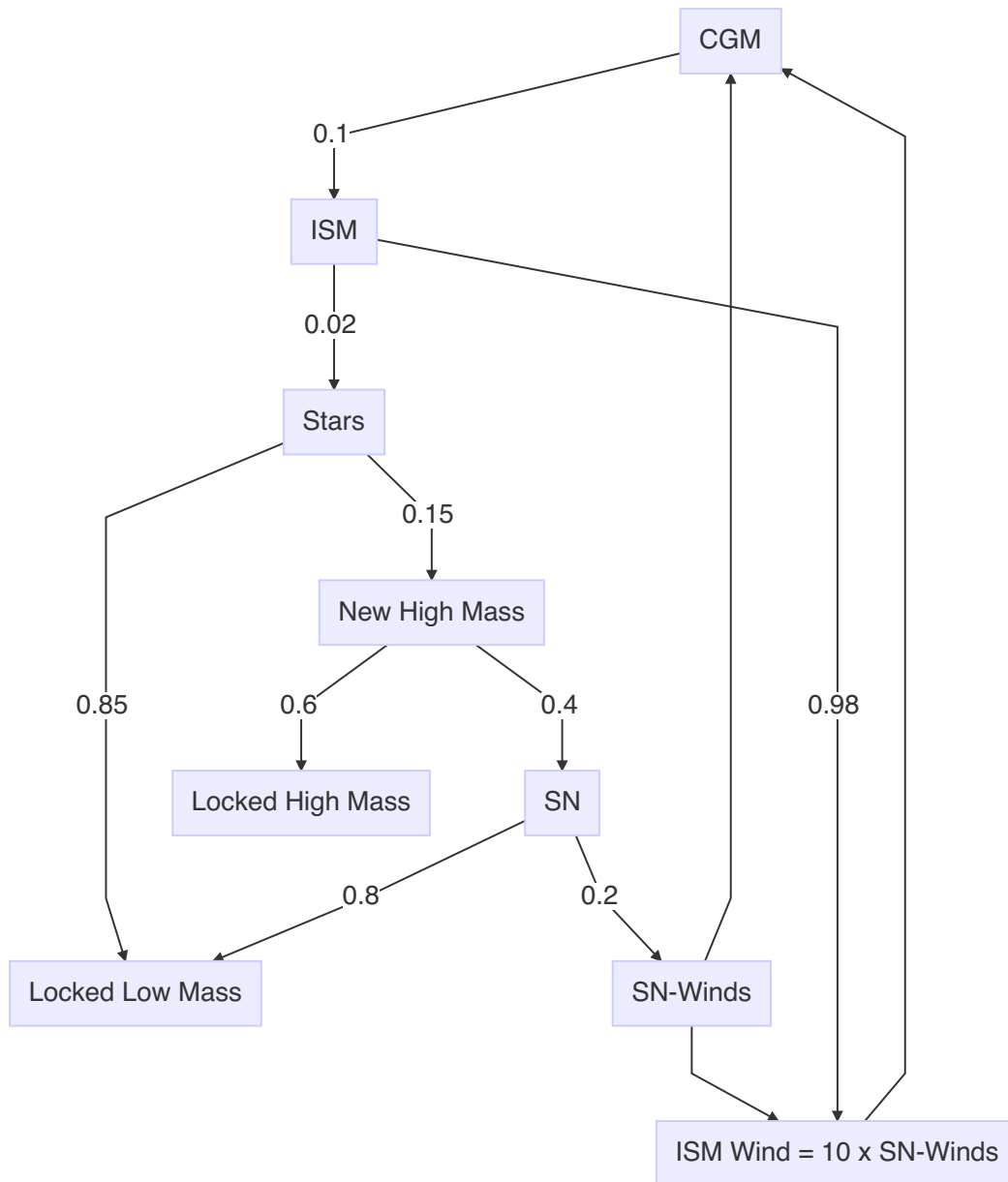
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```
##
## Attaching package: 'sphereplot'
## The following objects are masked from 'package:celestial':
##
##      car2sph, sph2car
```

1) [10 marks total]

Consider the simple open boxed model of star formation below, where each of the below cycles takes 10 Myrs:

```
mermaid("
graph TB
    CGM --> |0.1| ISM[ISM]
    ISM --> |0.02| Stars[Stars]
    Stars --> |0.85| LowMass[Locked Low Mass]
    Stars --> |0.15| NewHighMass[New High Mass]
    NewHighMass --> |0.4| SN[SN]
    NewHighMass --> |0.6| LockHighMass[Locked High Mass]
    SN --> |0.8| LowMass
    SN --> |0.2| SN-Winds[SN-Winds]
    SN-Winds --> ISM-Winds
    SN-Winds --> CGM
    ISM --> |0.98| ISM-Winds[ISM Wind = 10 x SN-Winds]
    ISM-Winds --> CGM
")
```



CGM = circumgalactic medium (gas outside galaxy), ISM = interstellar medium (gas inside galaxy), SN = supernovae

Note on “ISM Wind”: The SNe also blows out ISM gas into the CGM: 10 times the SN-Wind mass is lost from the ISM into the CGM.

If the CGM initially contains all mass in the system, after how much time is at least 50% of mass is locked up in low and high mass stars?

2) [10 marks total]

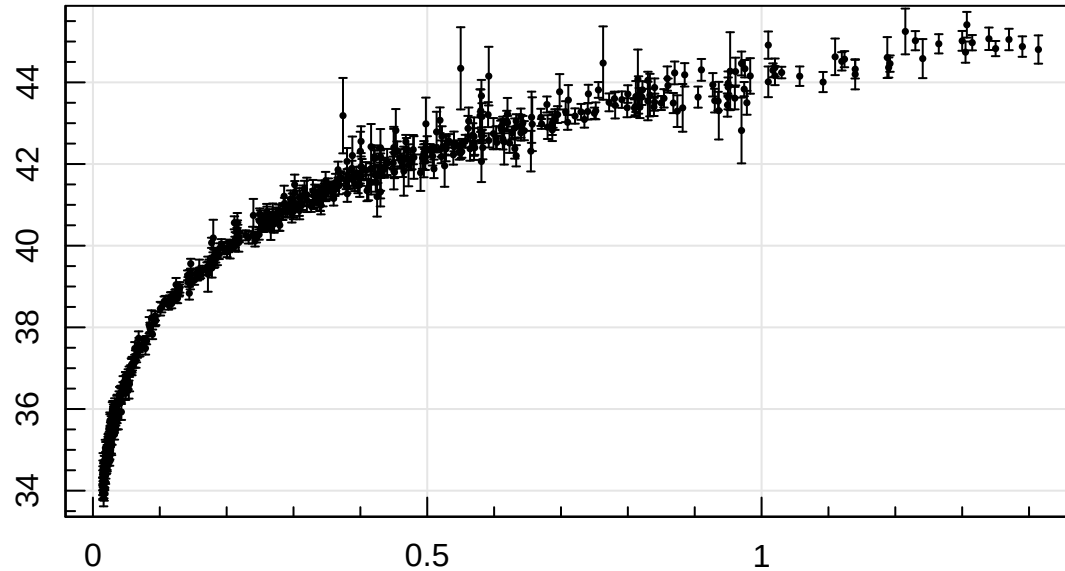
Get the Union 2.1 data from the *data* folder in the course DropBox. This is an ASCII table with tab-separated columns: Supernova Name, Redshift, Distance Modulus, and Distance Modulus Error. For Union2.1, there is an additional column for the probability that the supernova was hosted by a low-mass galaxy.

```
union = fread('../data/SCPUnion2.1_mu_vs_z.tab', skip=5) #this path will need to be appropriate f
#Don't worry about the column names warning, we fix that below:
```

```
names(union) = c('Name', 'Redshift', 'DistMod', 'DistMod_err', 'Prop')
```

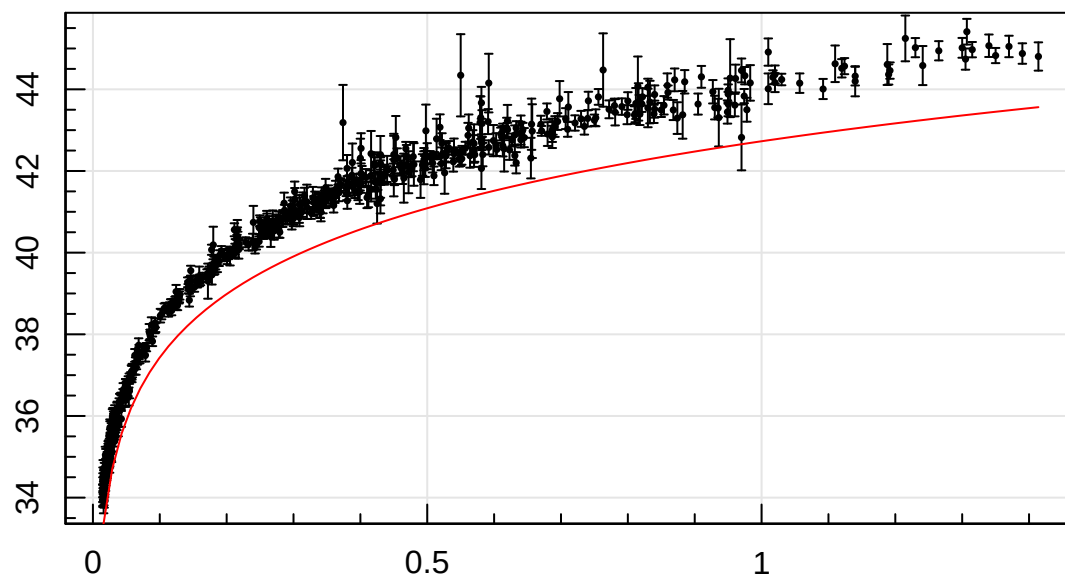
We can plot redshift against distance modulus.

```
magplot(union[,list(Redshift,DistMod)],pch=16,cex=0.5,grid=TRUE)
magerr(union$Redshift, union$DistMod, ylo=union$DistMod_err)
```



In the **celestial** package (you might need to install this) we have a function **cosdistDistMod** that translates between redshift and distance modulus. E.g.

```
magplot(union[,list(Redshift,DistMod)], pch=16, cex=0.5, grid=TRUE)
magerr(union$Redshift, union$DistMod, ylo=union$DistMod_err)
curve(cosdistDistMod(x, H0=100, OmegaM=1, OmegaL=0), add=TRUE, col='red')
```



Create an appropriate likelihood function based on Normal statistics (i.e. the likelihood is computed using the sum of per-observation likelihoods).

- a) [4] What is the value of this function at exactly $H_0 = 75$, $\Omega_M = 0.5$ and $\Omega_L = 0.5$?
- b) [3] Find the value of H_0 , Ω_M and Ω_L that maximises the likelihood of the observed redshift vs distance modulus data from Union 2.1. Assuming you do a good job, you have just recreated the analysis that won the 2011 Nobel Prize (which included Australian Brian Schmidt, the recent Vice Chancellor of ANU). Note if you do not supply fitting limits to your evaluation you

might get un-physical solutions that throw errors. I suggest starting with the parameters used in part a) i.e. $H_0 = 75$, $\Omega_M = 0.5$ and $\Omega_L = 0.5$.

c) [3] What are the marginalised ‘1-sigma’ errors on each of these parameters? (Use the Hessian).

3) [10 marks total]

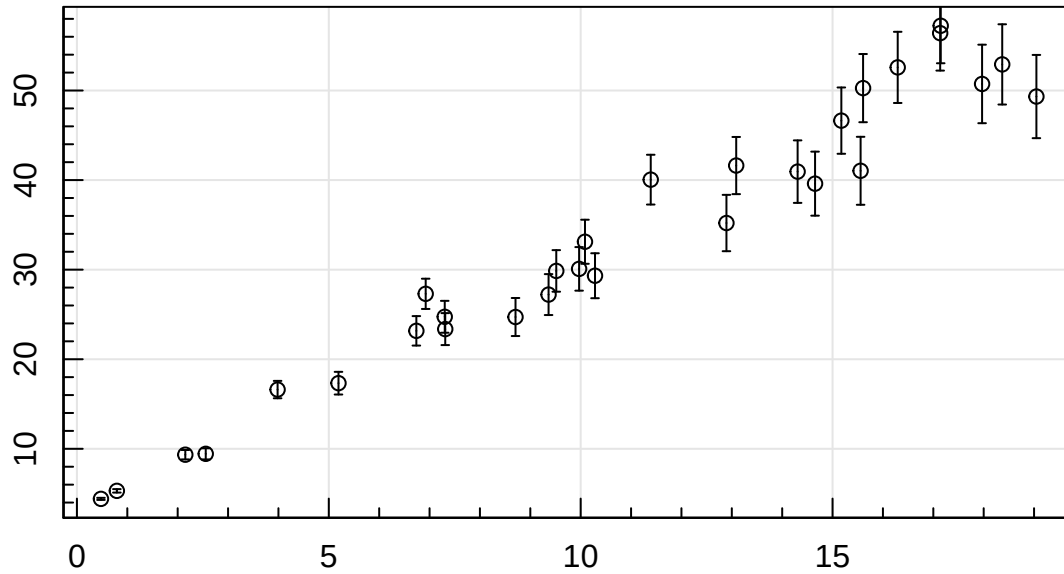
Below is some linearly correlated data with known Normal error on y :

assign5_q3

```
##          x          y    y_sigma
##      <num>    <num>    <num>
##  1:  0.4767131  4.412275  0.1162715
##  2:  0.7925006  5.308922  0.1932928
##  3:  2.1515870  9.343628  0.5247773
##  4:  2.5570536  9.441060  0.6236716
##  5:  3.9831201 16.612511  0.9714927
##  6:  5.1922592 17.328600  1.2664047
##  7:  6.7378154 23.171764  1.6433696
##  8:  6.9232901 27.302440  1.6886073
##  9:  7.3012055 24.739216  1.7807818
## 10:  7.3114990 23.365604  1.7832924
## 11:  8.7063311 24.714312  2.1234954
## 12:  9.3624575 27.219060  2.2835262
## 13:  9.5152286 29.860341  2.3207875
## 14:  9.9686451 30.087960  2.4313769
## 15: 10.0850432 33.115610  2.4597666
## 16: 10.2843171 29.323108  2.5083700
## 17: 11.3906211 40.049704  2.7782003
## 18: 12.8934416 35.206501  3.1447419
## 19: 13.0871506 41.623514  3.1919879
## 20: 14.3063384 40.944615  3.4893508
## 21: 14.6524031 39.603012  3.5737569
## 22: 15.1741524 46.637785  3.7010128
## 23: 15.5567464 41.036601  3.7943284
## 24: 15.6056905 50.266790  3.8062660
## 25: 16.2931063 52.581245  3.9739284
## 26: 17.1364963 56.403394  4.1796332
## 27: 17.1468631 57.226660  4.1821617
## 28: 17.9680060 50.732383  4.3824405
## 29: 18.3680455 52.915245  4.4800111
## 30: 19.0465296 49.326616  4.6454950
##          x          y    y_sigma
```

We can see how the known error in y grows with x :

```
magplot(assign5_q3[,list(x,y)], grid=TRUE)
mager(x=assign5_q3$x, y=assign5_q3$y, ylo=assign5_q3$y_sigma)
```



- [6] Given the data, what are the most likely values for the gradient (α) and offset (β) using the Normal distribution likelihood (i.e. the likelihood is computed using the sum of per-observation likelihoods)?
- [2] What are the marginalised '1-sigma' errors on the gradient (α) and offset (β)? (Use the Hessian).
- [2] What is the covariance and correlation between the gradient (α) and offset (β)? (Use the Hessian).

For this question you will encode the likelihood directly and optimise/sample it (do not use another package, e.g. **hyper.fit**).

4) [20 marks total]

You are presented with the results of an experiment. Our experimenter sampled (with replacement) 10^6 coins from a bag with a fraction of coins (F_b) that are biased and will come up heads with a probability P_b (the rest are unbiased). To make our lives even harder they tossed each coin a number of times determined from a Poisson distribution with an unknown λ . All they recorded was the number of heads they recorded each time. And made a table of the results:

```
##          n          x
##    <int>  <int>
##  1:      0  21549
##  2:      1  76790
##  3:      2 141259
##  4:      3 174601
##  5:      4 167767
##  6:      5 138265
##  7:      6 102126
##  8:      7  69502
##  9:      8  45544
## 10:      9  28374
## 11:     10  16581
## 12:     11   9124
## 13:     12   4630
## 14:     13   2285
## 15:     14    953
## 16:     15    400
## 17:     16    150
## 18:     17     69
## 19:     18     19
```

```
## 20:    19      8
## 21:    20      4
## 22:    21      0
## 23:    22      0
##      n      x
```

In the above n refers to the number of heads observed in a given trial, and x the number of times this number of heads (n) was observed (so the frequency). The only known limit is that $\lambda < 20$. Using uninformative Uniform priors, what are the most likely values for F_b , P_b and λ ?

- a) [5] To get you started, here is a partial version of the likelihood function you need to form (up to a parameter independent scaling factor). Fill in the correct symbols where there are ‘?’:

$$L(F_b, P_b, \lambda) \propto \prod_{i=1}^{23} \left(\sum_{N=0}^{\infty} [?.dbinom(k=?_i, n=N, p=?)+?.dbinom(k=?_i, n=N, p=0.5)].dpois(k=N, \lambda=?)^{?_i} \right),$$

where i is the element number of our frequency table, n_i is the value of the frequency bin in our table (how many heads we observed, $n_i = i - 1$ $i = 1 \dots 23$ for our data), N is the number of coins tossed based on our Poisson sample, and x_i is the number of times we observed that specific number of heads in our frequency table. F_b , P_b and λ all have the meanings specified above. For **dbinom**, k is the number of observations, n is the sample size and p is the probability of a successful observation.

- b) [3] Code up the correct likelihood in **R**
- c) [3] Compute the log-likelihood when $F_b = 0.2$, $P_b = 0.7$ and $\lambda = 8$.
- d) [4] Compute an **optim** maximum likelihood solution starting with the initial conditions from part c).
- e) [5] Using the **optim** solution as the initial starting point, run a useful number of MCMC steps of the above likelihood function with **LaplacesDemon** and produce a triangle plot of how the posterior parameters behave (consider using **magtri** to make this easier). This will take a few minutes, so probably keep it out of your Rmarkdown until you are happy it is working and add right at the end during final compilation.